

Math 111: Calculus Practice Exam 1

For Exam 1, in class on Friday, September 25

What Exam 1 is like. Four problems in a fifty-minute period, closed-book and device-free, covering everything through the tangent line approximation (§1.1, §1.3–§1.8). It is built so that a well-prepared student finishes in about forty minutes. Each problem is marked holistically on the 0–5 scale: a right answer with no reasoning is not a 5, and a wrong answer containing a real idea is not a 0. No differentiation rules are needed anywhere — everything is done from the definition or from reasoning about the picture.

How to use this. Sit down with a blank sheet and forty minutes on a clock, book closed and phone in another room. Doing that once is worth more than reading solutions three times. Bring whatever you could not finish to review on Wednesday the 23rd.

One note: problem 4 uses the tangent line approximation, which we cover on Monday the 21st. Save it until then.

1. The definition, by hand

about 7 minutes

Let $f(x) = x^2 - 3x$.

- (a) Use the limit definition to compute $f'(2)$. Show the algebra, and do not skip the step where the h cancels.
- (b) Use the definition again to find $f'(a)$ for an arbitrary a , and check that it agrees with part (a).
- (c) In one sentence, say what $f'(2)$ tells you about the graph of f at $x = 2$.

2. Everything from one picture

about 11 minutes

The graph of a function f on $[0, 7]$ is shown. No formula is given.



- (a) Sketch f' on $[0, 7]$ on axes of your own, marking clearly any place where f' does not exist.
- (b) At which x in $(0, 7)$ does f' fail to exist? Say precisely what goes wrong, and say whether f is continuous there.
- (c) On which subintervals of $[0, 7]$ is f' increasing? Justify from the graph of f itself, not from your sketch in part (a).
- (d) Put $f'(1)$, $f'(3)$, and $f'(5)$ in increasing order, and justify from the picture.

3. A reservoir, and what the numbers are worth

about 11 minutes

Let $V(t)$ be the volume of water in a reservoir, in millions of gallons, t days after September 1.

t (days)	0	3	6	9	12
$V(t)$ (millions of gallons)	4.0	4.9	5.5	5.8	5.9

- (a) Estimate $V'(6)$. State the units, and write one sentence saying what your estimate means to somebody who manages the reservoir.
- (b) Give an estimate of $V'(6)$ that is certainly too large and one that is certainly too small. Say which is which, and what you are assuming about V' in order to be sure.
- (c) Is V' increasing or decreasing over these twelve days? What does your answer say about the reservoir?

4. The tangent line as an approximation

about 11 minutes

This problem uses §1.8, which we cover on Monday, September 21.

Suppose f satisfies $f(4) = 12$ and $f'(4) = -0.8$, and f is concave down everywhere near $x = 4$.

- (a) Write down the tangent line to f at $x = 4$ and use it to estimate $f(4.5)$.
- (b) Is your estimate too large or too small? Justify using the concavity, and draw a picture that makes the argument visible.
- (c) Would you use the same method to estimate $f(9)$? Explain what does and does not go wrong, and say what you would need to know to trust such an estimate.